



Large Language Models in Ground Mobility

Innovations and Challenges

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Created by ChatGPT according to the title

What are LLMs

LLMs are advanced **AI** models trained on **massive** datasets of **text** to understand and generate **human-like language**

Core Functions

- Text comprehension
- Natural conversation (chatbots)
- Content creation (articles, summaries)
- Code generation
- Translation



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<https://mergingminds.bureauworks.com/episode/when-words-have-no-root-a-conversation-with-lisa-meeden>

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Large Models vs. Large Language Models

Large Models: large ML models, including various deep learning models like CNNs and RNNs.

Large Language Models: large models used in **Natural Language Processing (NLP)**, primarily for understanding and generating text. Examples include GPT and BERT.

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LLMs vs. Multimodal LLMs

Multimodal Large Language Models: advanced AI models that integrate and process multiple types of data, such as **text, images, and audio** to understand and generate human-like responses, while the language component remains central to how the model processes and produces information.

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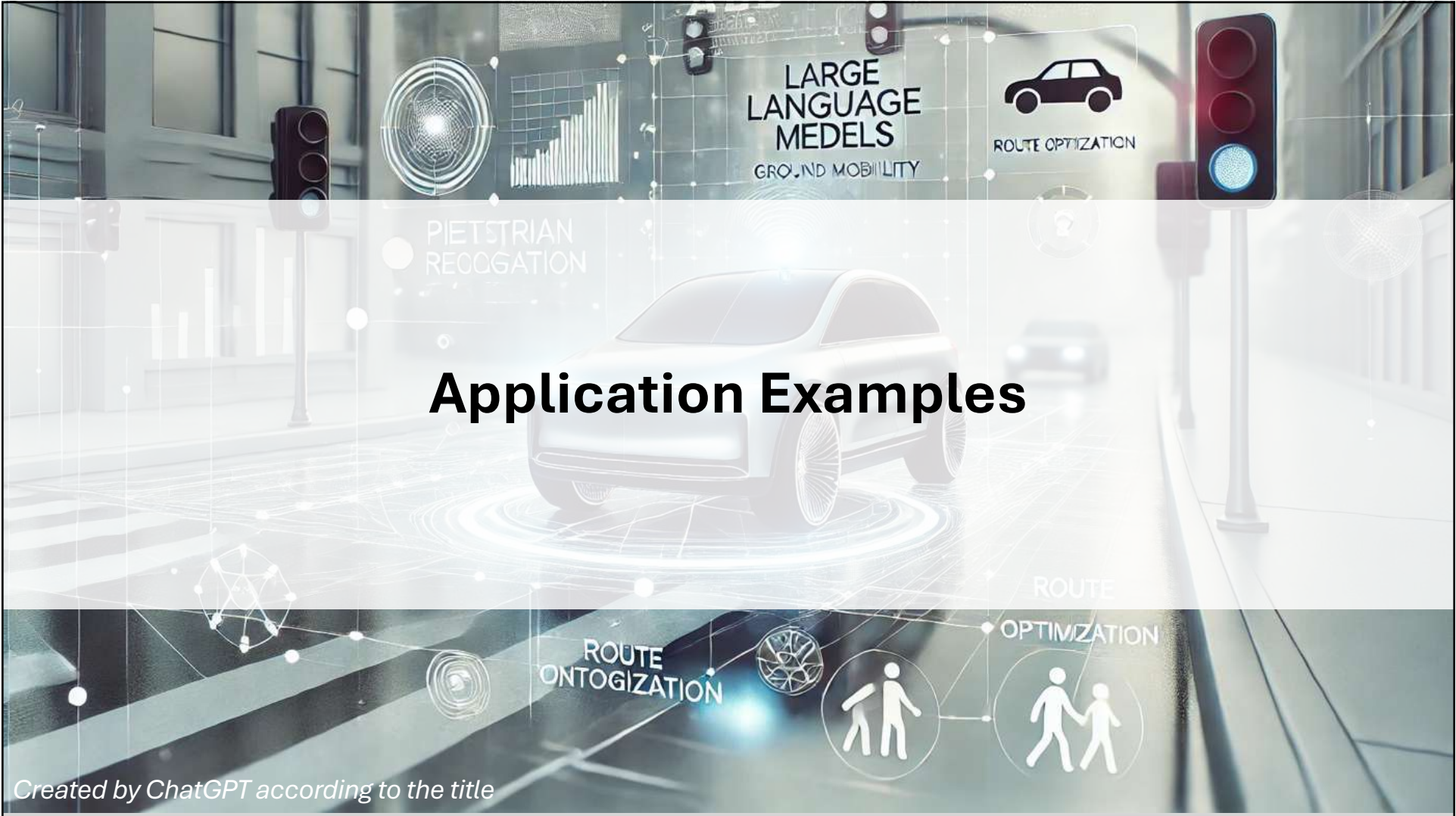


Unstructured Data

- Image Understanding
- Visual Question Answering
- Image Generation
- Audio Processing

} Multimodal LLMs





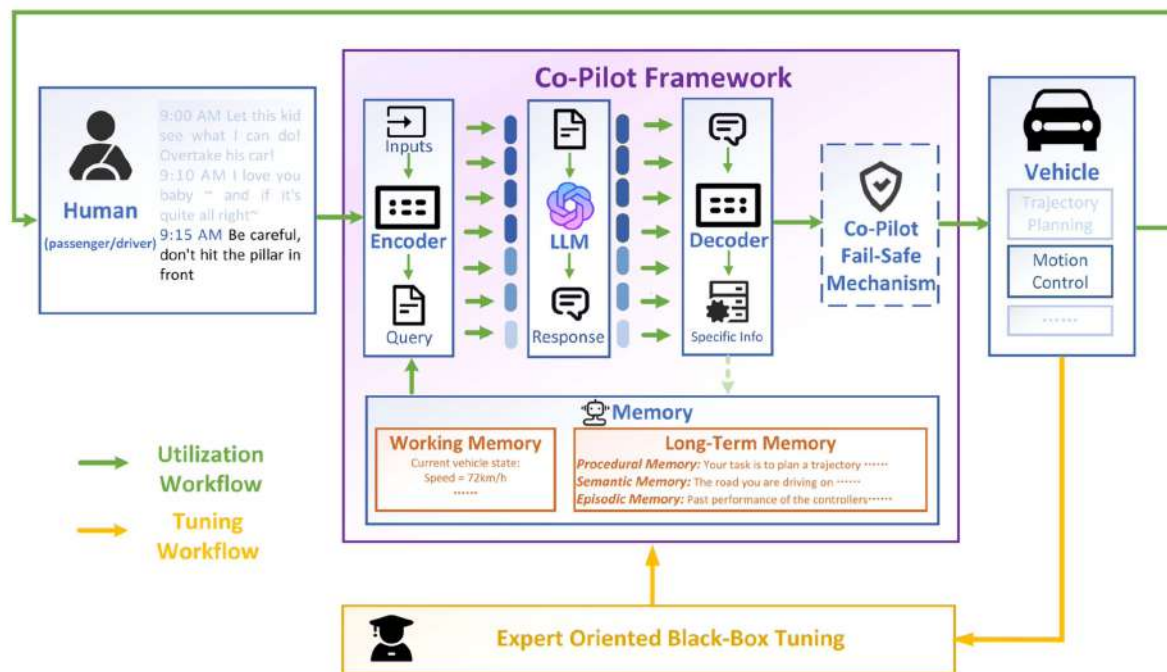
The image depicts a futuristic urban environment with a central car. Overlaid on this scene are several AI application examples, each represented by an icon and a label. In the upper left, there is a radar icon labeled 'PIETSTRIAN RECCOGATION' (sic). To its right is a bar chart icon labeled 'LARGE LANGUAGE MEDELS' (sic) with 'GROUND MOBILITY' below it. Further right is a car icon labeled 'ROUTE OPTIZATION'. In the lower left, there is a network diagram icon labeled 'ROUTE ONTOGIZATION' (sic). In the lower right, there is a pedestrian icon labeled 'ROUTE OPTIMIZATION' (sic). The background features traffic lights and buildings, creating a sense of a smart city.

Application Examples

Created by ChatGPT according to the title

1. Lower barrier between users and outcomes

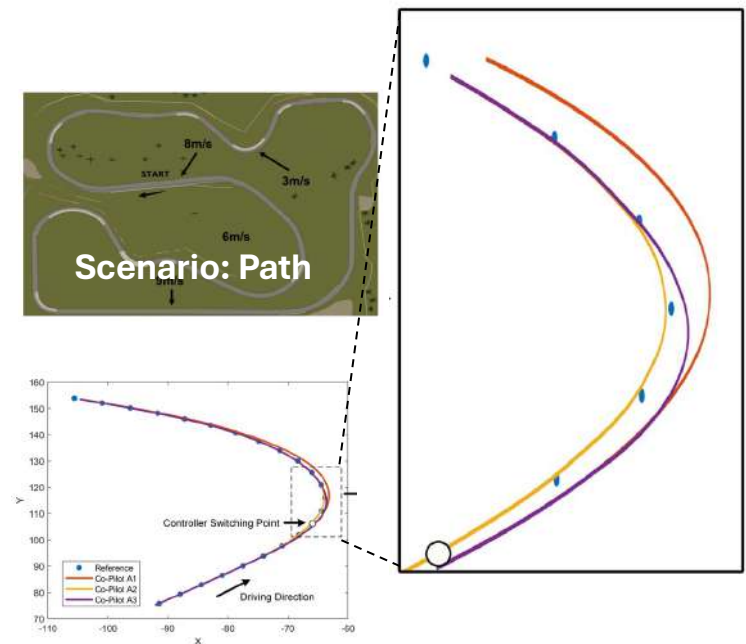
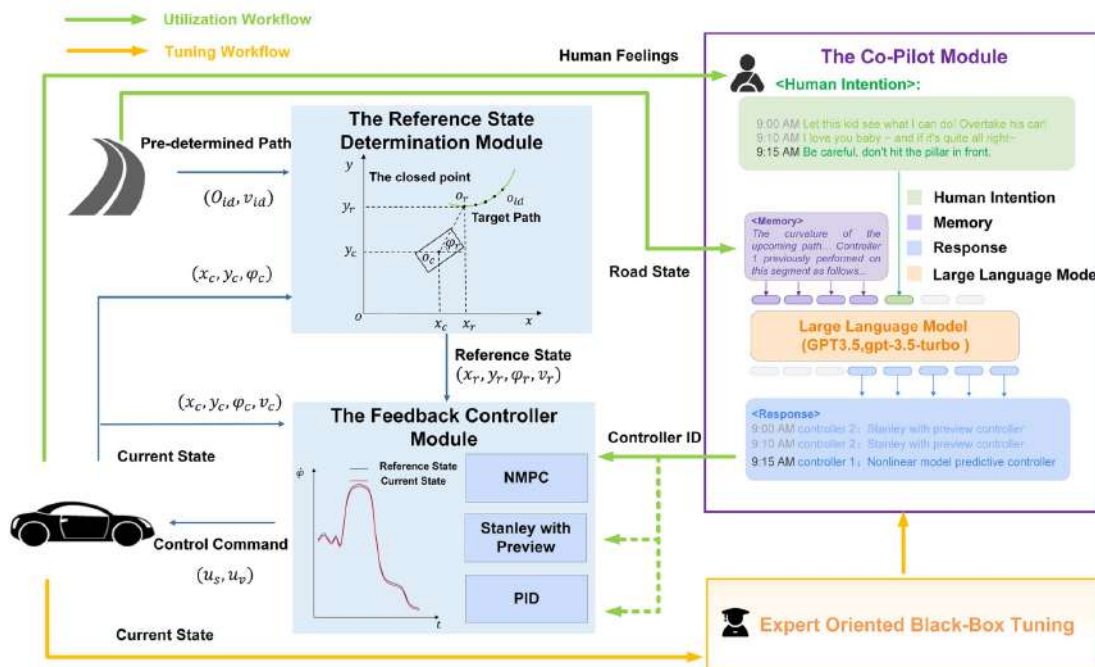
ChatGPT as Your Vehicle Co-Pilot: An Initial Attempt



One of the most challenging problems in human-machine co-work is **the gap between human intention and the machine's understanding and execution.**

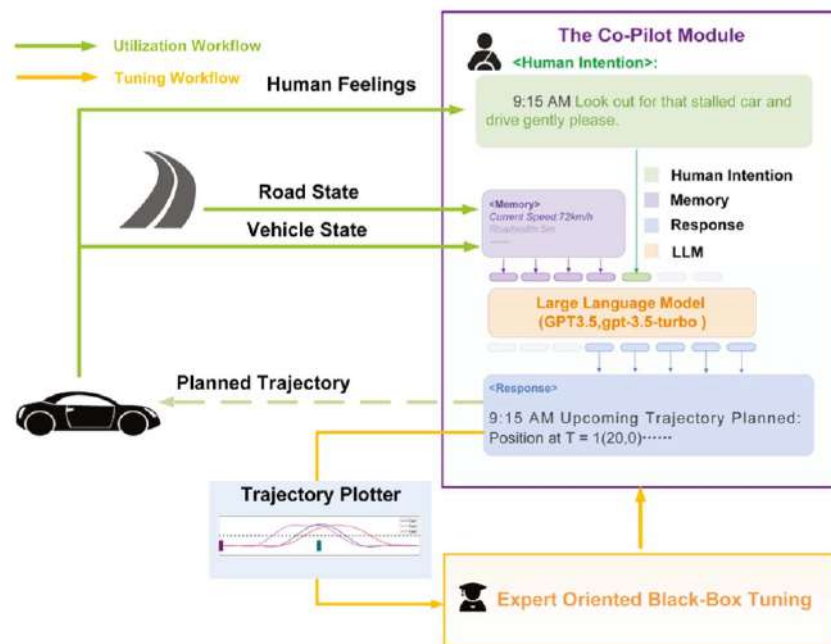
1. Lower barrier between users and outcomes

Co-Pilot: Test 1 - Path tracking control

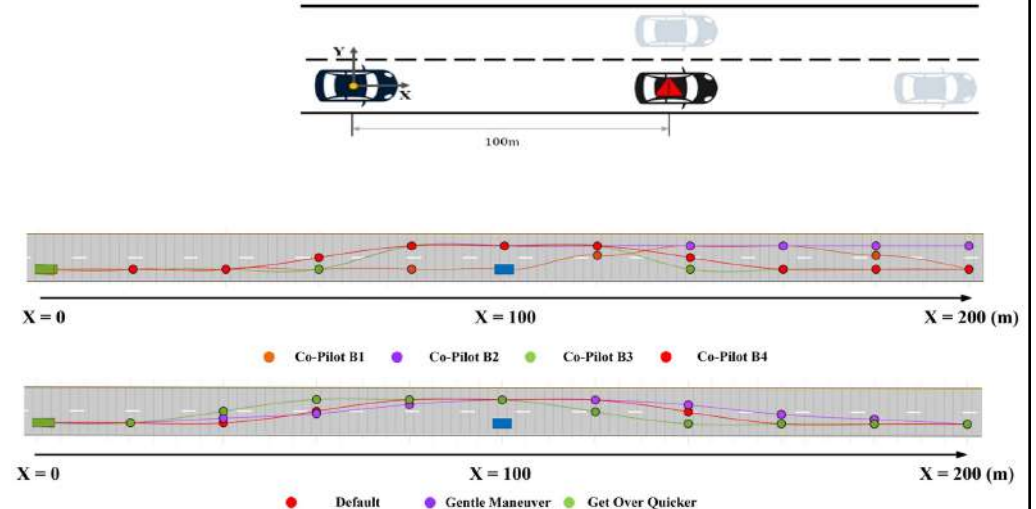


1. Lower barrier between users and outcomes

Co-Pilot: Test 2 - Trajectory planning

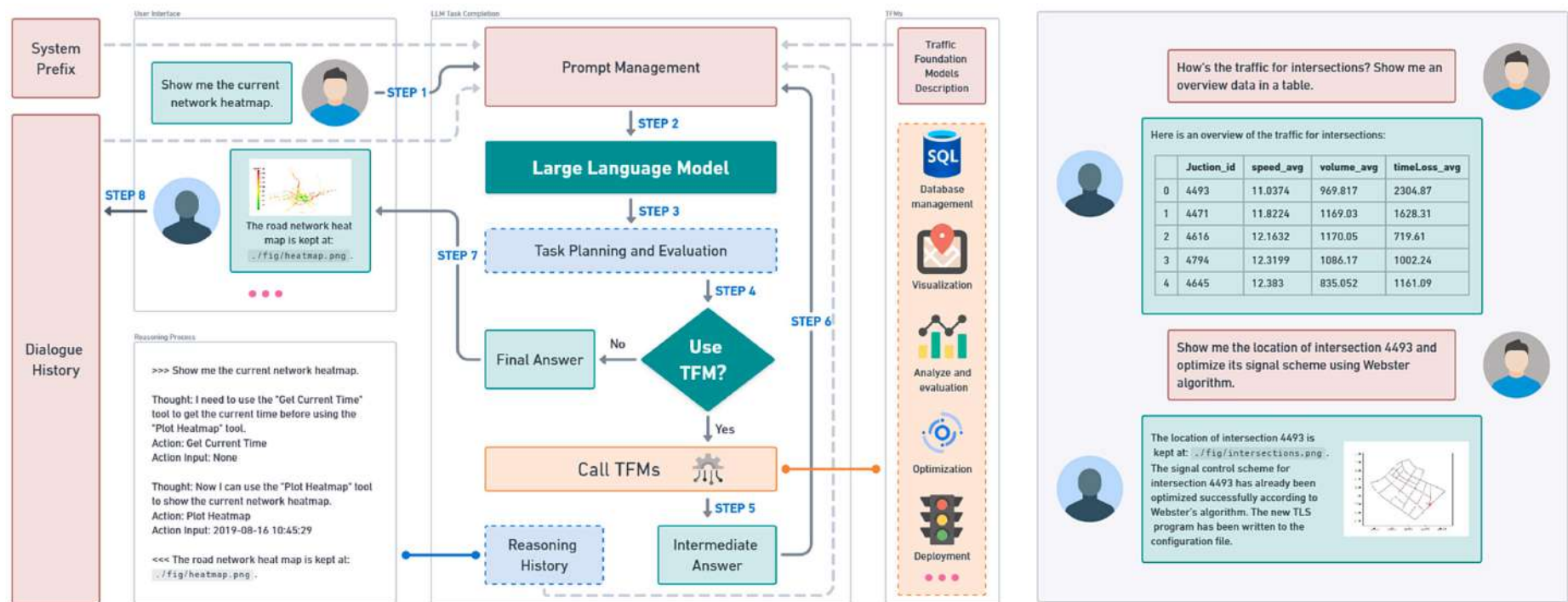


Scenario: Overtaking



1. Lower barrier between users and outcomes

TrafficGPT: Viewing, processing, and interacting with various traffic models (functions)



TrafficGPT: Viewing, processing and interacting with traffic foundation models

www.GoTrafficGo.com

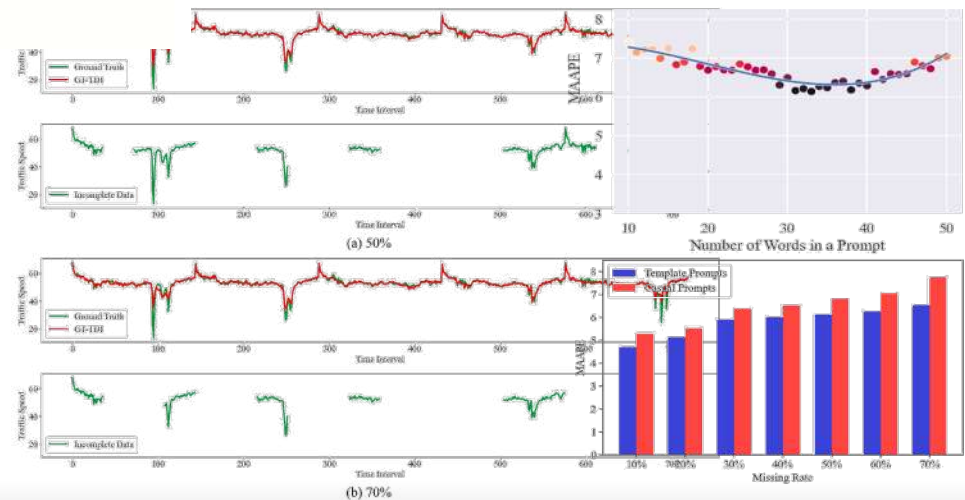
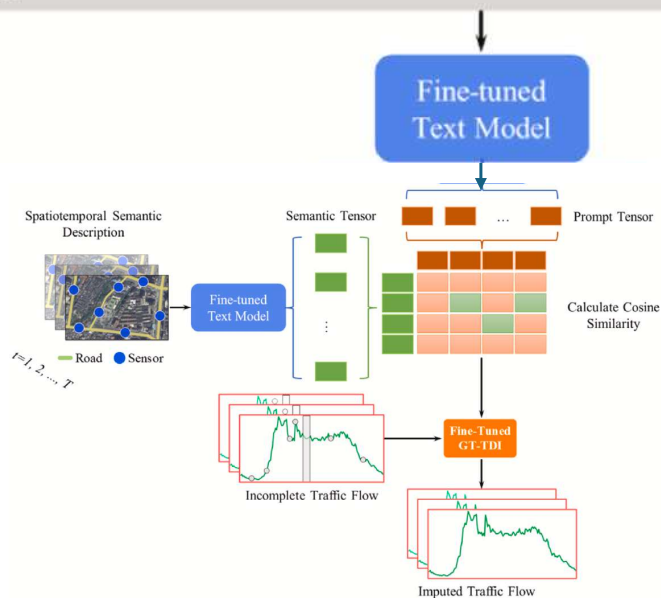
1. Lower barrier between users and outcomes: Prompt Engineering

Prompts

1. What was the traffic flow on the San Diego Freeway nearby Los Angeles International Airport on July 3rd, 2013, between 6 am and 12 am?
2. Cleanse the traffic flow data collected from the San Diego Freeway in proximity to Los Angeles International Airport during the month of May 2013.

...

***Prompt engineering** allows for designing specialized prompts or modifying model inputs to guide the behavior of LLMs*



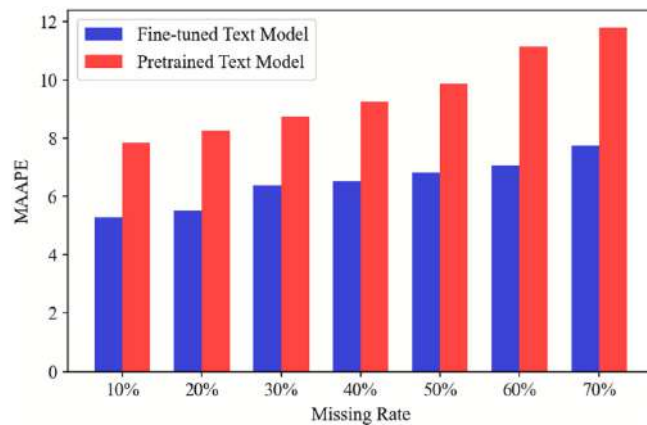
Semantic understanding and prompt engineering for large-scale traffic data imputation

www.GoTrafficGo.com

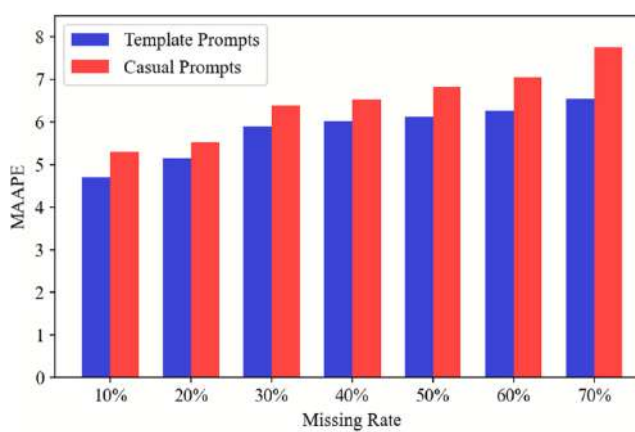
1. Lower barrier between users and outcomes: Prompt Engineering

Effect of Prompt

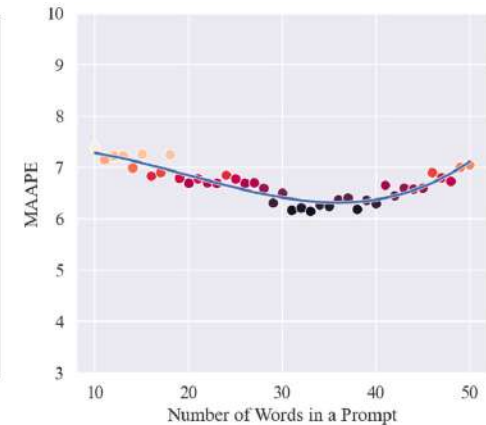
Necessity of fine-tuned



Effect of Prompt template



Effect of Prompt length



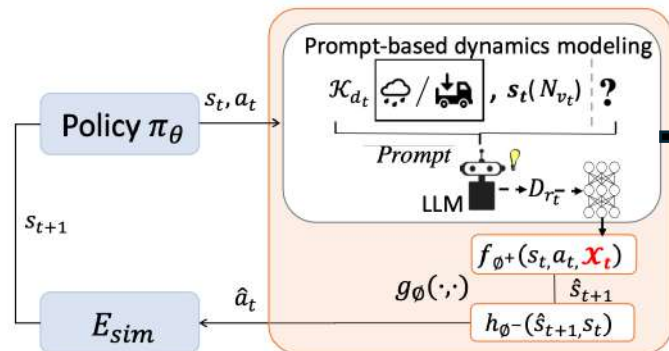
1. Lower barrier between users and outcomes



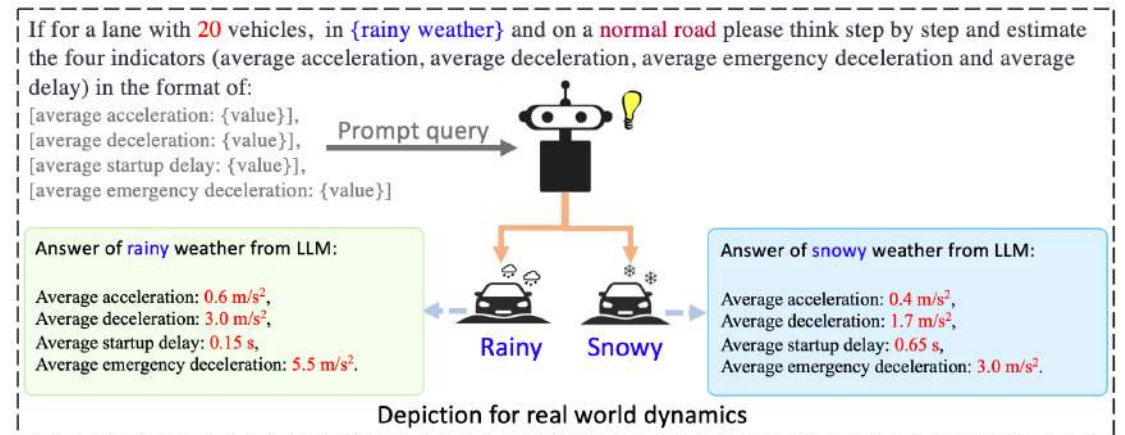
Lower the barrier between users and outcomes, by interpreting users' demands and selecting models

2. Help model adapt to real-world settings

PromptGAT uses LLMs to generate realistic dynamic information, improving Sim-to-Real Transfer in traffic signal control and enhancing performance in real-world environments

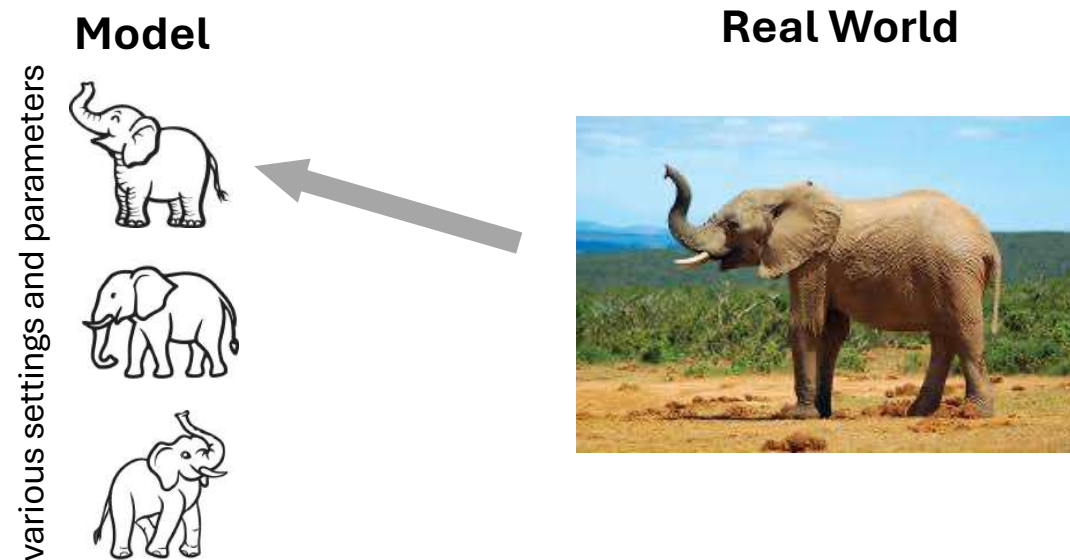


Grounded Action Transformation (RL)



LLM

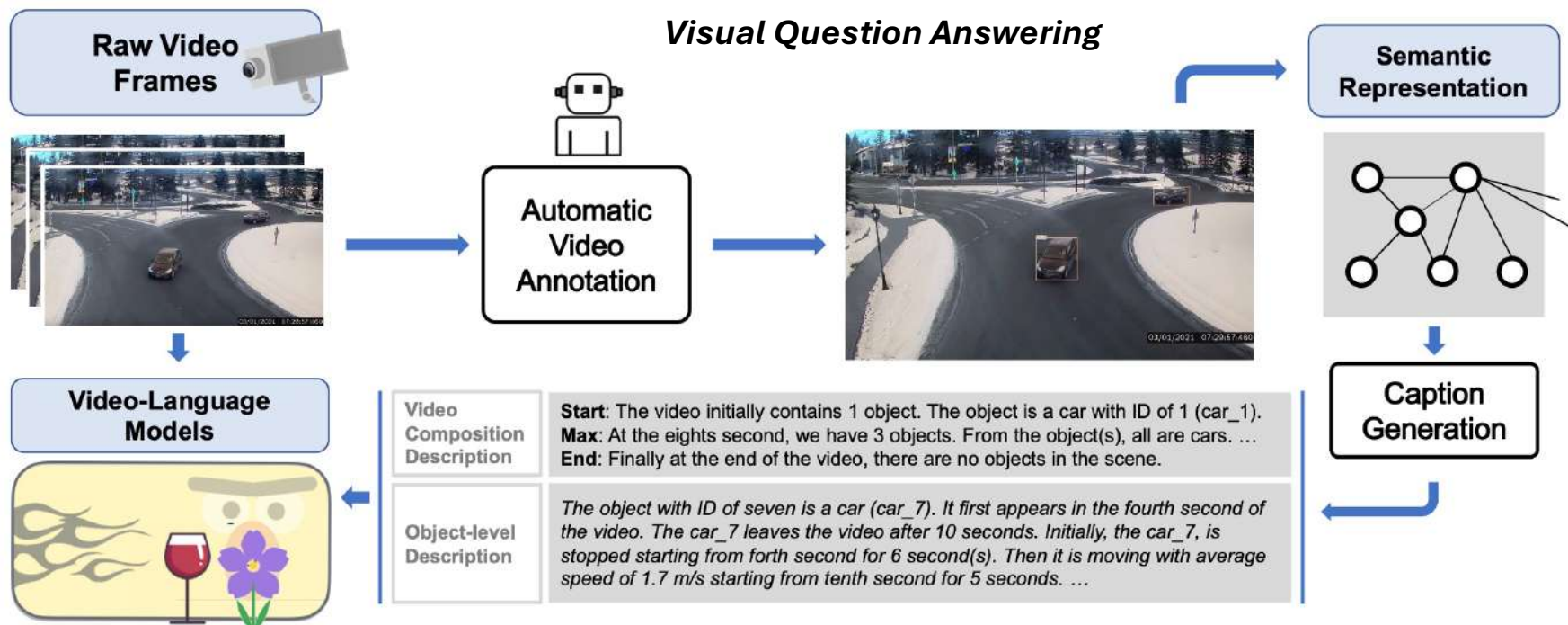
2. Help model adapt to real-world settings



Help model better adapt to real-world settings, which are usually complicated and tedious for human

3. Understand unstructured data: traffic (monitoring) videos

Just like a tireless human



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Just like a tireless human



- Traffic Accidents
- Traffic Congestion
- Pedestrian Crossing
- Road Obstructions
- Illegal Parking
- Vehicle Breakdown
- Pedestrian Accidents
- Emergency Vehicle Passage
- Traffic Signal Violations
- Unusual Vehicle Behavior
- Weather-Related Events
- Road Condition Deterioration
- Queueing or Bottlenecks
- Unusual Traffic Patterns

4. Reduce burden of summarization: Text mining

Just like a tireless human

Tweet	Detected topic (MetRoBERTa)	Sentiment	Service	Route/station/car	Gender
"@Metrorailinfo What's happening with the green line between West Hyattsville and PG Plaza for the last twenty odd minutes?"	Operation/Delay/Procedures	Neutral	Rail	Green/West Hyattsville, Prince George's Plaza	Male
"@wmata train car 7507 needs to be cleaned. It has some makeup or something on the inside. It's in the glass and seat. Ruined a nice woman's clothes."	Cleanliness/Maintenance	Negative	Rail	Car 7507	Female
"@Metrorailinfo Operator of Yellow line train to Huntington (Archives station at 8:49 am, containing car #3200) was a de-light this morning. Welcoming aboard and announcing each station clearly and with infectious joy."	Customer service	Positive	Rail	Yellow/archives	Male
"@WeMoveDC @wmata I live in Fairlawn and it's been so nice to be able to take a 30 bus from Pennsylvania Ave to 4th and Independence SW and it only take 10 minutes. There should be dedicated bus lanes there"	Operation/Delay/Procedures	Positive	Bus	Route 30/4th and Independence SW	Male
"Hi @wmata the driver of the 70 Metrobus #6032 needs a re-fresher on safe driving. He cut me	Unsafe driving/Road conflict	Negative	Bus	Route 70/ vehicle 6032	Male

Tweet	Detected topic (MetRoBERTa)	Sentiment	Service	Route/station/car	Gender
"@wmata train car 7507 needs to be cleaned. It has some makeup or something on the inside. It's in the glass and seat. Ruined a nice woman's clothes."	Cleanliness/Maintenance	Negative	Rail	Car 7507	Female
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more likely erroneous information from the bus tracking system."					
"Hope to see more transit agencies connect #micromobility services with transit offerings. @DavidZipper - love this deep dive. PWNV @wmata's Bus Transformation Project makes this recommendation (s/o to the innovative thinking of @travelin-gail)"	General	Positive	NA	NA	Male

Sample Tweets Classification

Main Functions

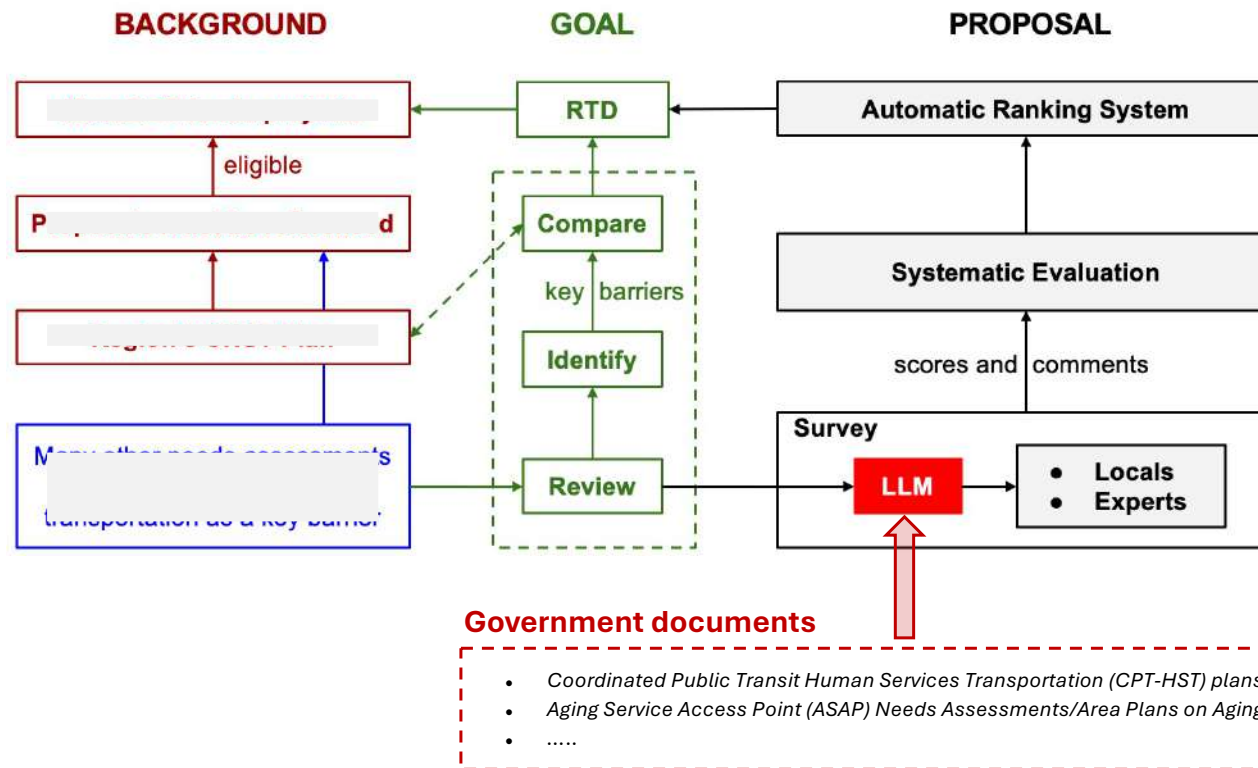
- Adding Structure to Unstructured Data
- Understanding Sources of Customer Feedback
- Generating Performance Reporting Metrics from Customer Feedback

MetRoBERTa: Leveraging traditional ... data to develop a transit-topic-aware language model

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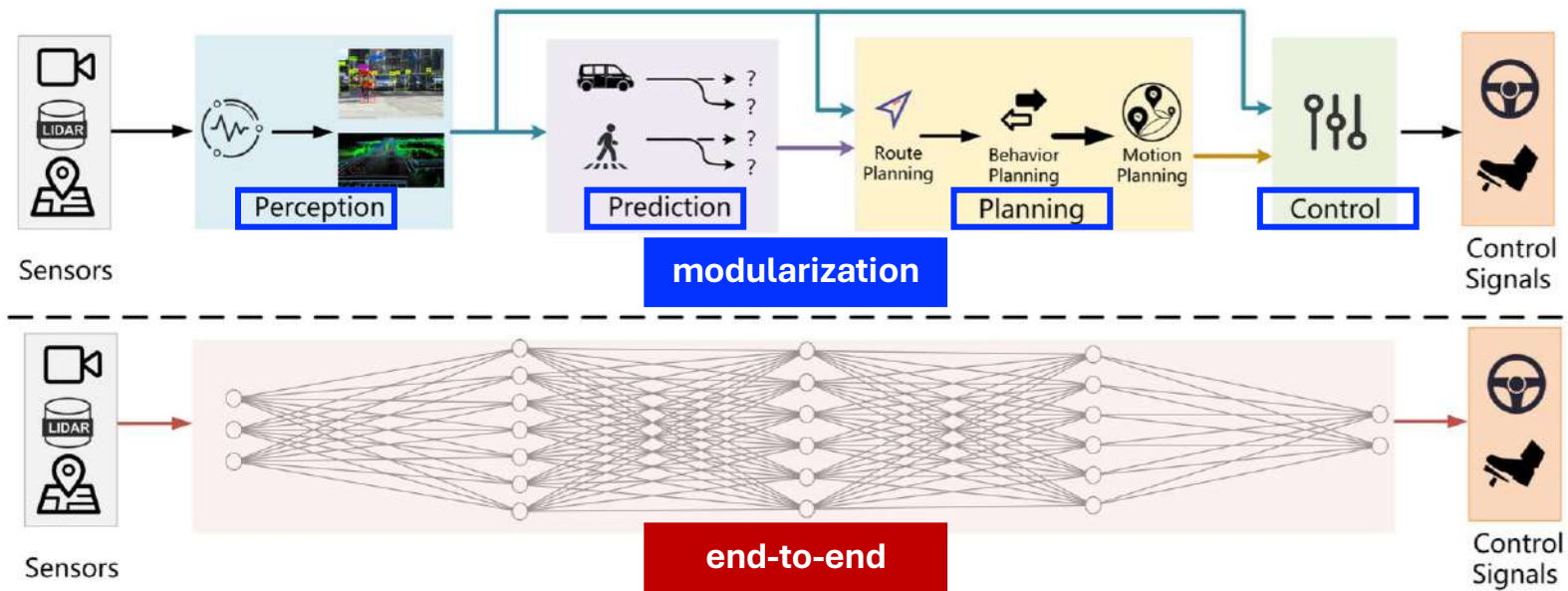
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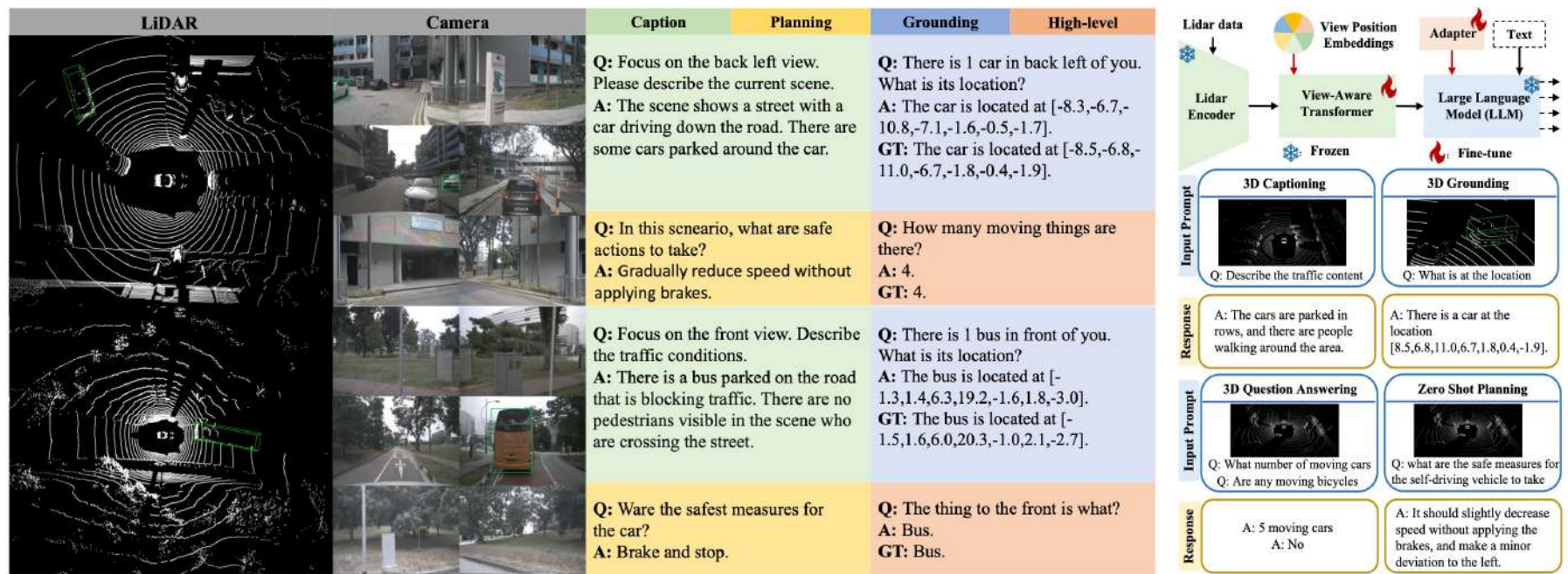
5. Empower Autonomous Driving

AV roadmap: Modularization vs. End-to-end



5. Empower Autonomous Driving

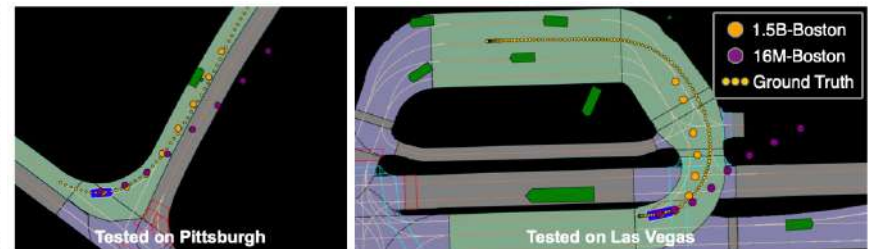
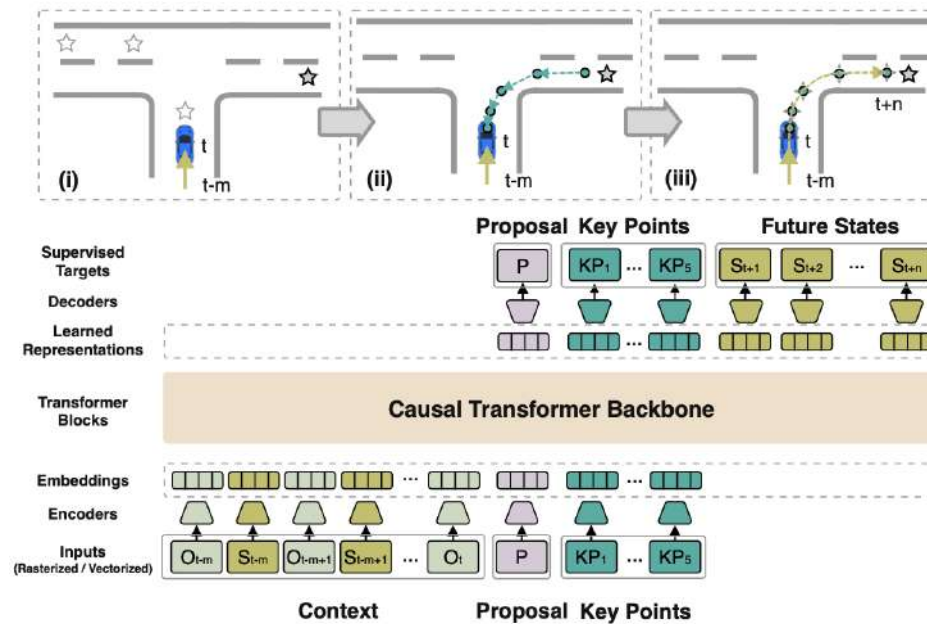
Modularization: Perception



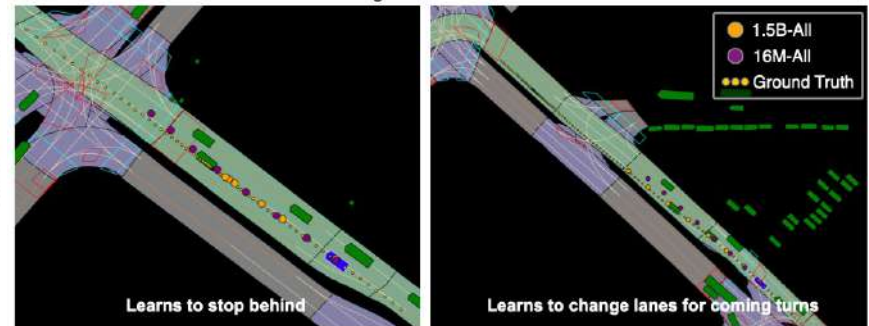
Integrating Language-Derived Appearance Elements with Visual Cues in Pedestrian Detection

5. Empower Autonomous Driving

Modularization: Prediction



Planning Tested on Different Cities

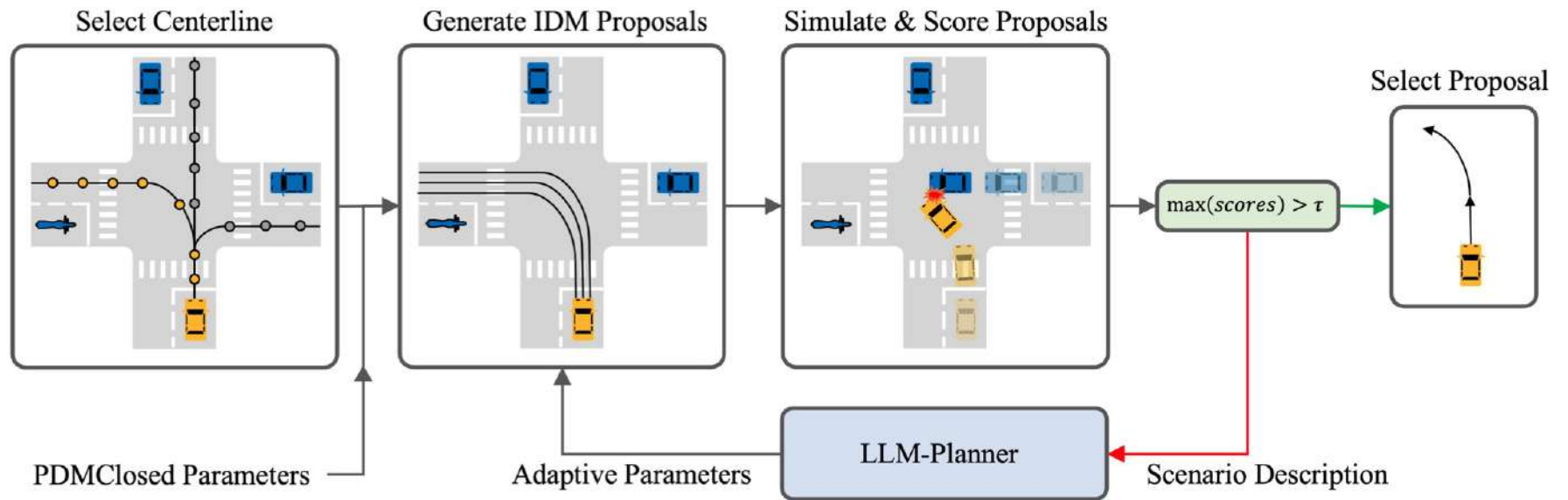


Emerging Semantics Learning

Large trajectory models are scalable motion predictors and planners

5. Empower Autonomous Driving

Modularization: Planning

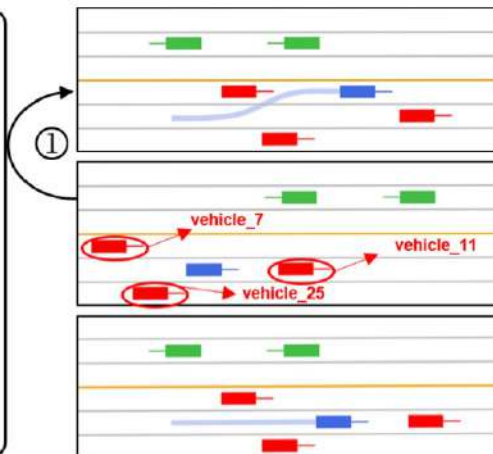


LLM-ASSIST: Enhancing Closed-Loop Planning with Language-Based Reasoning

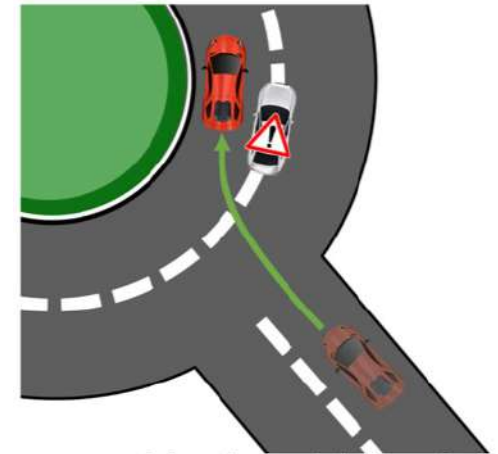
5. Empower Autonomous Driving

Modularization: Control

User: ... Please drive **AGGRESSIVELY** while staying safe.
LLM: ... vehicle_7, vehicle_25, and vehicle_11 may affect the movement of the ego vehicle. vehicle_11 in front of me is slow, and vehicle_7 behind me on the left is **relatively far away and slow**. Out of the premise of aggressive driving, I should **overtake from the left**.



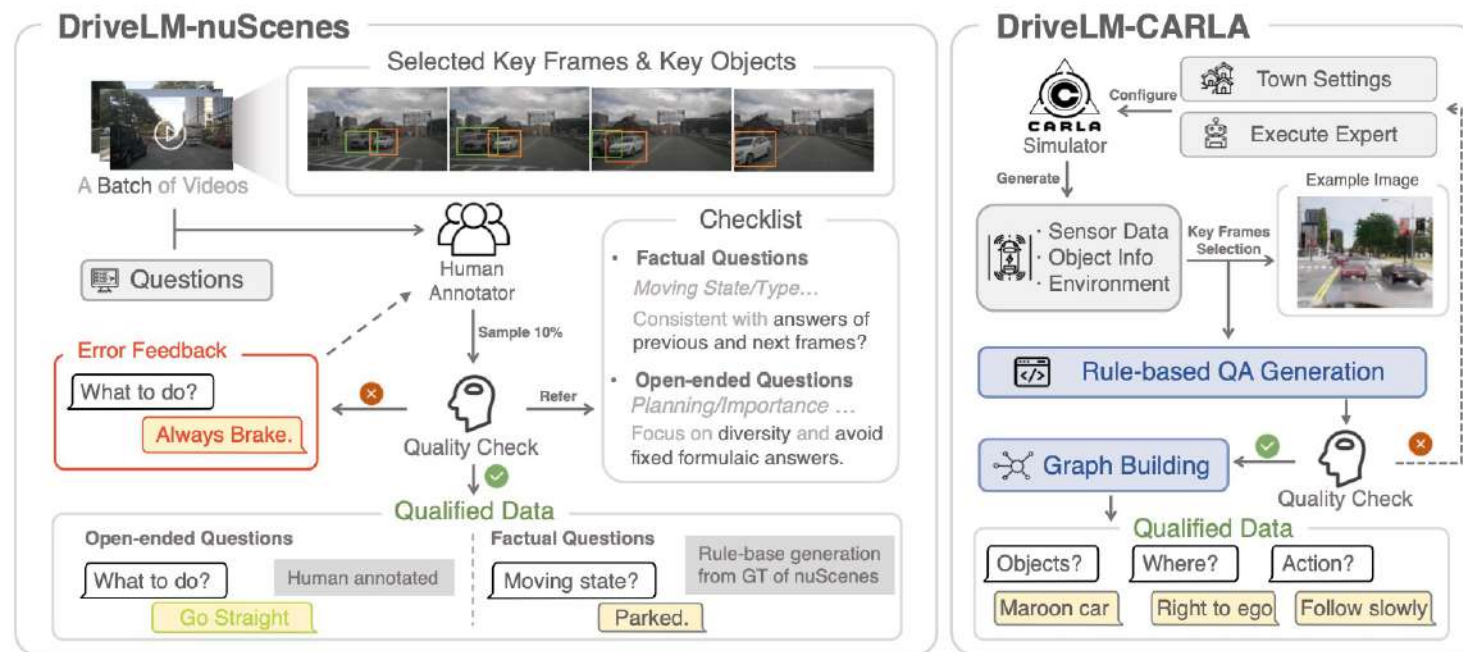
User: ... Please drive **CONSERVATIVELY**.
LLM: ... vehicle_7, vehicle_25, and vehicle_11 may affect the movement of the ego vehicle. vehicle_11 in front of me is slow, but vehicle_7 and vehicle_25 behind me **may affect my overtaking**. Out of the premise of conservative driving, I should **slow down and follow the vehicle**.



LANGUAGE MPC: Large Language Models as Decision Makers for Autonomous Driving

5. Empower Autonomous Driving

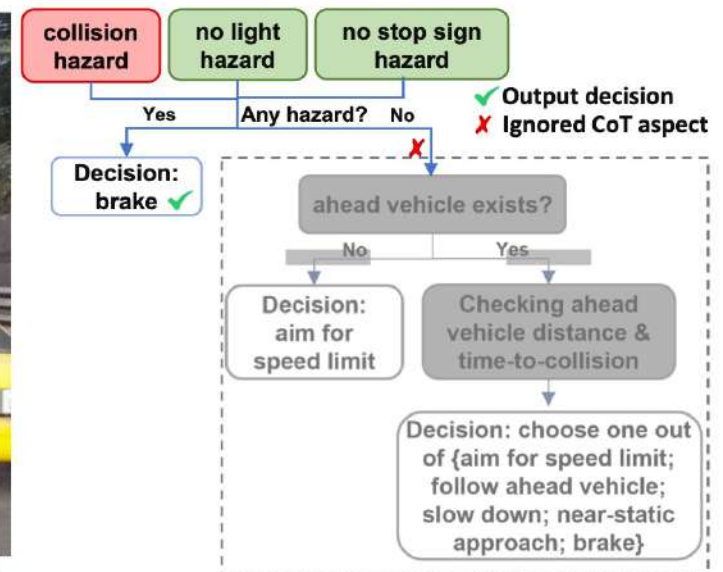
End-to-end: Data fundamentals



DriveLM: Driving with Graph Visual Question Answering

5. Empower Autonomous Driving

End-to-end



DriveCoT: Integrating Chain-of-Thought Reasoning with End-to-End Driving



The image depicts a futuristic urban environment with a car at the center. The scene is overlaid with a network of white lines and nodes, suggesting a data-driven or AI-powered system. Various icons and text labels are scattered throughout, representing different aspects of mobility and AI. The labels include 'LARGE LANGUAGE MEDELS' (sic), 'GROUND MOBILITY', 'ROUTE OPTIMIZATION', 'PIETSTRIAN RECCOGATION' (sic), 'ROUTE ONTOGIZATION' (sic), and 'ROUTE OPTIMIZATION'. The car is a sleek, modern vehicle with a glowing blue light on its front. The background shows a city street with buildings, traffic lights, and a crosswalk. The overall aesthetic is high-tech and futuristic.

Limitation

Created by ChatGPT according to the title

Limitation: Reproducibility issue



Generated
by
ChatGPT

Key factors that contribute to reproducibility issues in LLMs

- Randomness in Training and Inference
- Lack of Standardized Environments
- Model Size and Resource Constraints
- Model and Data Availability
- Lack of Transparent Reporting

Possible **Solutions** to Improve Reproducibility:

- Standardized Reporting
- Environment Reproducibility Tools
- Reproducible Research Practices

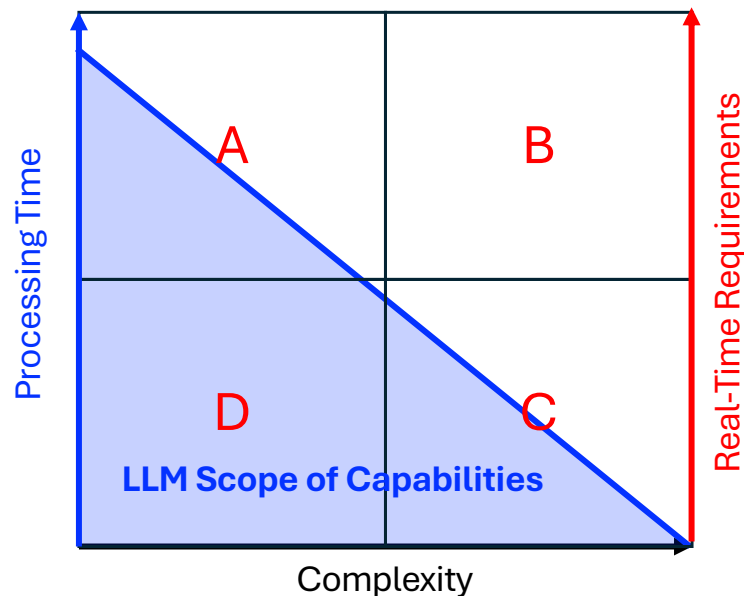


Limitation: Lack of domain knowledge



While LLMs have a broad understanding of many topics due to their training on diverse sources like books, articles, and websites, **they may not possess the specialized, in-depth knowledge required for highly technical fields** such as medicine, law, or engineering (transportation). In these areas, the models might struggle with accuracy or nuanced reasoning. However, their domain knowledge can be **improved through fine-tuning** on domain-specific datasets or integrating external knowledge sources like specialized databases or expert systems.

Limitation: Processing speed



A. Low Complexity, High Real-Time Requirements

- Traffic Signal Control
- Traffic Accident Detection and Response
- Vehicle Speed Monitoring and Violation Detection
- Road Occupancy Detection

B. High Complexity, High Real-Time Requirements

- Autonomous Vehicle Decision Making
- Real-Time Traffic Event Detection and Response
- Vehicle-to-Everything (V2X) Communication and Coordination

C. High Complexity, Low Real-Time Requirements

- Long-Term Optimization of Intelligent Traffic Systems
- Traffic Pattern and Behavior Prediction
- Traffic Network Design and Planning
- Traffic Accident Risk Analysis and Prevention

D. Low Complexity, Low Real-Time Requirements

- Traffic Data Collection and Reporting
- Traffic Infrastructure Maintenance Scheduling
- Road Safety Audits
- Public Transportation Route Optimization (long-term planning)

Limitation: “Hallucination” problem



"Hallucination" in LLMs refers to the generation of results that are inaccurate or **do not align with the facts or user requirements**, often without a sufficient factual basis.

Limitation: Lack of understanding of physical world



Unlike humans or robots, LLMs lack the ability to interact with or learn directly from the environment. They do not have a true understanding of physical concepts like gravity, momentum, etc., as their knowledge is **entirely based on the text they were trained on.**

Limitation: Privacy and security



LLMs raise significant privacy and security concerns, particularly **when trained on sensitive data** that could unintentionally expose personal information.



LARGE
LANGUAGE
MEDELS
GROUND MOBILITY

ROUTE OPTIZATION

PIETSTRAN
RECCGATION

Thank you

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he.zb@hotmail.com | hezb@mit.edu

ROUTE
OPTIMIZATION

ROUTE
ONTOGIZATION

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